

Morphology and phylogeny of scalopine moles (Eulipotyphla: Talpidae: Scalopini) from the eastern Himalayas, with descriptions of a new genus and species

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All scalopine moles are found in North America, except the Gansu mole (*Scapanulus oweni*), which is endemic to central-west China. In 2019, we collected two specimens of Scalopini on Mt Namjagbarwa in the eastern Himalayas, Tibet, China. We sequenced two mitochondrial (*CYT B* and 12S) and three nuclear (*APOB*, *BRCA1* and *RAG2*) genes to estimate the phylogenetic relationships of the two moles, and also compared their morphology with other genera and species within the Scalopini. Both morphological and molecular analyses strongly suggest that the specimens represent a new monotypic genus and species, which are formally described here as *Alpiscaptulus medogensis* **gen. et sp. nov.** The dental formula of the new mole (44 teeth) is distinct from the Chinese *Scapanulus oweni* (36 teeth) and its hairy and pale brown tail is unique among species of the Scalopini. The Kimura-2-parameter (K2P) distances of *CYT B* between *A. medogensis* and the four recognized Scalopini genera range from 14.5% to 18.9%. A sister relationship between *A. medogensis* and *Scapanulus oweni* was strongly supported in the phylogenetic trees. The divergence between *A. medogensis* and *Scapanulus oweni* occurred in the mid-Miocene (c. 11.56 Mya), which corresponds with the rapid uplift of the Himalayan-Tibetan Plateau.

ADDITIONAL KEYWORDS: China – Mammalia – taxonomy.

INTRODUCTION

The family Talpidae has diverse lifestyles, including terrestrial, semi-fossorial, semi-aquatic, aquatic-fossorial and fully fossorial. The Scalopini is one of the two fully fossorial mole tribes within the family Talpidae. Currently, six species within four genera (*Parascalops* True, 1894, *Scalopus* É. Geoffroy, 1803, *Scapanulus* Thomas, 1912 and *Scapanus* Pomel, 1848) are recognized in the Scalopini tribe (Hutterer, 2005; Wilson & Mittermeier, 2018). Within this tribe, most genera are monotypic and only *Scapanus*

contains three species [*Scapanus latimanus* (Bachman, 1842), *Scapanus orarius* (True, 1896) and *Scapanus townsendii* (Bachman, 1839)]. All extant scalopines are found in North America, except the Gansu mole (*Scapanulus oweni* Thomas, 1912), which is restricted to a limited area in central-west China, including Gansu, Qinghai, Shaanxi, Sichuan, Chongqing and Hubei (Wilson & Mittermeier, 2018).

The macroevolution of the Scalopini is more complex than generally acknowledged. Although the extant scalopines are now sparse and mainly distributed in North America, more than 14 fossil genera from the upper Oligocene have been found throughout Asia, Europe and North America, indicating a formerly high diversity and wide distribution (Schwermann *et al.*, 2019). Morphological studies suggest that the Scalopini originated in Eurasia and had at least two back

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migrations to Eurasia (Sánchez-Villagra *et al.*, 2006; Schwermann *et al.*, 2019). The extant Gansu mole (*Scapanulus oweni*) was regarded as a migrator and the only relict species of the Scalopini in China. Because of its limited distribution, little is known about the general biogeographical history of the Scalopini in China.

The eastern Himalayas is one of the most important biodiversity hotspots in the world (Myers *et al.*, 2000) and is considered as a refuge during glaciations (Lei *et al.*, 2014). However, due to the extensive elevational gradient, harsh environment and logistical difficulties, a good understanding of its biological diversity remains elusive, especially in some areas of Tibet (Li *et al.*, 2020). In April–June 2019, we conducted a mammal survey along elevational gradients (650–3800 m a.s.l.) on Mt Namjagbarwa (29°44' N, 95°40' E; Fig. 1) in the eastern Himalayas, Medog County, Tibet, China. During the field survey, two specimens of a talpid mole were collected. The enlarged anterior-first incisors of the two specimens (Fig. 2) suggested that they were

members of the Scalopini. Morphologically, they were similar to the Gansu mole, but their pale brown tail and the possession of 44 teeth were obviously different from this species, indicating that they potentially represented a distinct taxon (hereafter, Scalopini sp.).

In this study, we obtained two mitochondrial and three nuclear genes of Scalopini sp. specimens collected during our study and published sequences of six recognized species of the Scalopini. We also collated information on their morphometric characters. We used these data to elucidate the taxonomy, phylogeny, and diversification of Scalopini sp. moles collected from Mt Namjagbarwa.

MATERIAL AND METHODS

MORPHOLOGICAL DATA AND COMPARISONS

Two specimens of Scalopini sp. were collected during a biodiversity survey of Mt Namjagbarwa from April

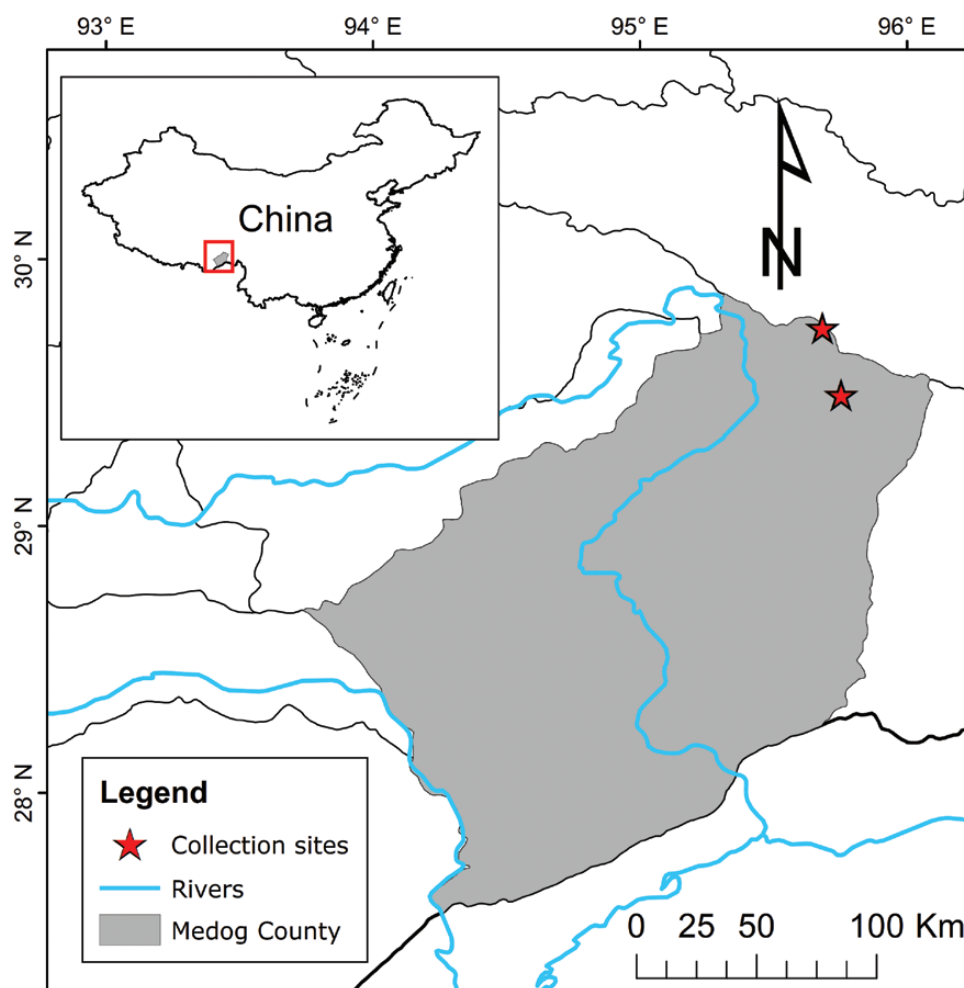


Figure 1. Map showing the collection sites of *Alpiscaptulus medogensis*.

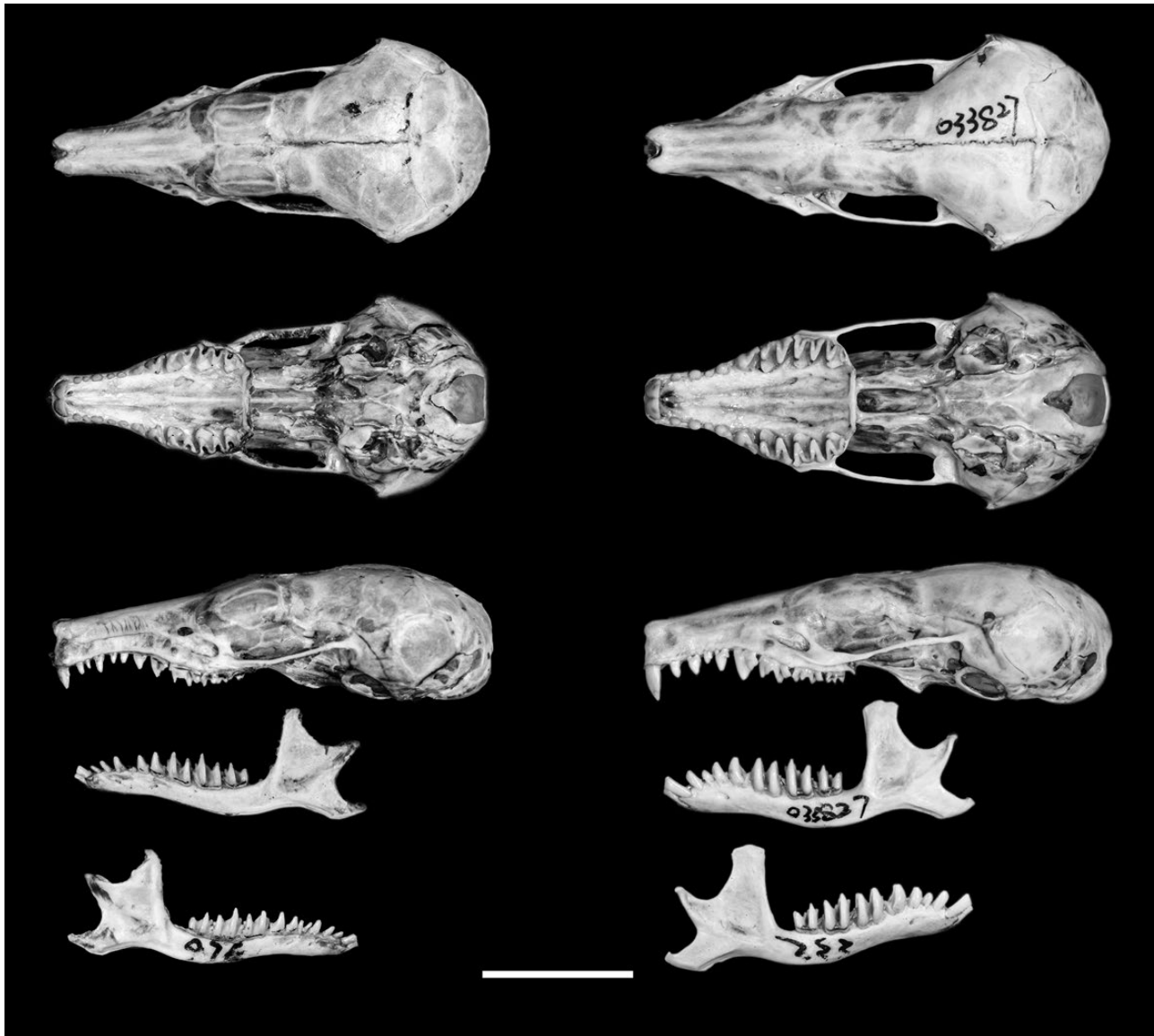


Figure 2. Dorsal, ventral and lateral views of the skull and lateral views of the mandible of the holotype of *Alpiscaptulus medogensis* (KIZ: 037966; left) and *Scapanulus oweni* (KIZ: 033872; right). Scale bar = 10 mm.

to June 2019. One specimen (KIZ: 037965; ♂) was collected at an elevation of 2400 m a.s.l., and the other (KIZ: 037966; ♀) was collected at 3700 m a.s.l. Both specimens were prepared as standard museum specimens, the skulls were cleaned, and the muscle and liver tissues were extracted and preserved in pure ethanol. The museum specimens and tissue samples were stored at Kunming Institute of Zoology (KIZ), Chinese Academy of Sciences (CAS). Animal handling followed the animal care and user guidelines of the American Society of Mammalogists (Sikes *et al.*, 2016), and with the permission of the Forestry and Grassland Administration of Xizang (Tibet) Autonomous Region and local authorities.

We also examined and measured the six specimens of *Scapanulus oweni* collected by the Laboratory of Mammal Ecology and Evolution at KIZ, from the Gansu, Shanxi and Qinghai provinces of China. For external morphological characterization, four external measurements, including head and body length (HB), tail length (TL), hind foot length (HF) and weight (W) were taken from specimen labels or field notes. For craniomandibular delineation, 15 variables were measured using a digital calliper calibrated to the nearest hundredth (0.01) millimetre following Pan *et al.* (2007) and Yang *et al.* (2007). The measurements included greatest length of the skull (GLS), basal length (BL), palatal length (PL), cranial height (CH),

cranial breadth (CB), interorbital breadth (IOB), zygomatic breadth (ZB), upper toothrow length (UTL), length of the upper molar row (UML), maximum width across the upper second molars (M^2 - M^2), distance from the front of upper fourth premolar to the end of upper third molar (P^4 - M^3), lower toothrow length (LTR), height of coronoid process (HCP), height of coronoid valley (HCV) and height of articular condyle (HAC).

Because one skull of *Scalopini* sp. and two skulls of *Scapanulus oweni* were partially damaged, only five intact skulls (one *Scalopini* sp. and four *Scapanulus oweni*) were available. Therefore, we did not perform statistical morphometric analyses. Instead, we directly compared *Scalopini* sp. morphology with other genera and species within the *Scalopini*. Comparative morphological characters of the scalopine species were obtained from Thomas (1912), Jackson (1915), Hallett (1978), Yates & Schmidly (1978), Hartman & Yates (1985), Carraway *et al.* (1993), Verts & Carraway (2001) and Elbroch (2006), and we followed their terminologies for consequent morphological descriptions.

MOLECULAR DATA AND ANALYSES

Total DNA was extracted using the Qiagen DNeasy Blood & Tissue Kit, following the manufacturer's instructions. We amplified two mitochondrial [cytochrome b (*CYT B*) and 12S rRNA] and segments of three nuclear genes [apolipoprotein B (*APOB*), breast cancer 1 (*BRCA1*) and recombination activating protein 2 (*RAG2*)]. These genes have been widely used in phylogenetic studies of moles. The primer sequences are outlined in Table 1. The PCR amplicons were sequenced in both directions on an ABI 3730 Genetic Analyzer (Applied Biosystems, Foster City, CA, USA). We downloaded corresponding sequences of six other scalopine species and four Chinese talpine moles [*Euroscaptor longirostris* (Milne-Edwards, 1870), *Euroscaptor parvidens* Miller, 1940, *Parascaptor leucura* (Blyth, 1850) and

Scaptonyx fuscicaudus Milne-Edwards, 1872] from GenBank (Table 2). We also obtained sequences of *Episoriculus caudatus* (Horsfield, 1851) and *Uropsilus nivatus* Allen, 1923 from GenBank to use as outgroups. Each gene was aligned using MUSCLE (Edgar, 2004). Analyses were performed on three concatenated datasets: (1) mitochondrial genes; (2) nuclear genes; and (3) mitochondrial-nuclear genes. The partitioning schemes and evolutionary models were estimated using PartitionFinder v.2.0 in PhyloSuite (Zhang *et al.*, 2020). The phylogenetic analyses were performed using maximum likelihood (ML) and Bayesian inference (BI) in IQ-TREE and MrBayes, respectively, in PhyloSuite. We considered the ultrafast bootstrap value (UFBoot) ≥ 95 and posterior probabilities [PP] ≥ 0.95 as strong supports (Huelsenbeck & Rannala, 2004; Minh *et al.*, 2018). We used BEAST v.1.8 (Drummond *et al.*, 2012) to estimate divergence times based on the Birth-Death model as the tree prior and relaxed lognormal as the clock model prior. Three calibrations were used to calibrate branch lengths: (1) *Geotrypus minor* Ziegler, 2012, which is regarded as the most ancient fossil taxon of the Talpini at 33.9 Mya (Islamoglu *et al.*, 2010; Ziegler, 2012), we established the prior using an exponential distribution with an offset of 33.9 and a mean of 11.3 ($M = 41.73$ Mya, 95% CI = 34.48–67.75 Mya); (2) *Scalopoides* sp. (UNSM Dw-121), the oldest known *Scalopini*, was estimated at 30.8 Mya (Bailey, 2004), we established an exponential distribution with offset of 30.8 and mean of 10.27 ($M = 37.92$ Mya, 95% CI = 31.33–61.57 Mya) for this calibration; (3) divergences of Chinese and North American scalopines were estimated in the mid-Miocene between 10 and 22 Myr (He *et al.*, 2016), we established a normal distribution ($M = 16$, $SD = 1.5$) for this calibration. Each analysis consisted of 100 000 000 generations with resampling every 10 000 generations. Convergence was assessed using TRACER v.1.7 (Rambaut *et al.*, 2018).

Table 1. Primers used for PCR and sequencing

Locus	Primer name	Primer sequences	Sense/anti-sense	Reference
<i>CYT B</i>	L14724_hk3	GGACTTATGACATGAAAAATCATCGTTG	Sense	He <i>et al.</i> , 2010
	H15915_hk3	GATTCCTCCATTTCTGGTTTACAAGAC	Anti-sense	He <i>et al.</i> , 2010
12S	L613_hk1	GGCGGGCGAGCAAAGCACTGAAAATG	Sense	He <i>et al.</i> , 2010
	H1478_hk1	TGATTGGTGGAGGGTGACGAGCGGTGTGT	Anti-sense	He <i>et al.</i> , 2010
<i>BRCA1</i>	B1f	TGAGAACAGCACTTTATTACTCAC	Sense	Dubey <i>et al.</i> , 2006
	B1r	ATTCTAGTTCCATATTGCTTATACTG	Anti-sense	Dubey <i>et al.</i> , 2006
<i>APOB</i>	ApoBf	GCAATCATTTGACTTAAAGTG	Sense	Dubey <i>et al.</i> , 2007
	ApoBr	GAGCAACAATATCTGATTGG	Anti-sense	Dubey <i>et al.</i> , 2007
<i>RAG2</i>	RAG2-F220	GATTCCTGCTAYCTYCCTCTCT	Sense	Teeling <i>et al.</i> , 2000
	RAG2-R995	CCCATGTTGCTTCCAAACCATA	Anti-sense	Teeling <i>et al.</i> , 2000

Table 2. Samples and sequences used for molecular analyses, new sequences generated in this study are shown in bold

Species	Museum code	Sample locality	CYT B	12S	APOB	BRCA1	RAG2
<i>Alpscapotulus medogensis</i>	KIZ: 037965	China: Tibet	MT349295	MT349878	MT349297	MT349299	MT349301
<i>Alpscapotulus medogensis</i>	KIZ: 037966	China: Tibet	MT349296	MT349879	MT349298	MT349300	MT349302
<i>Scapanulus oweni</i>	KIZ: 036922	China: Gansu	KX754480	KX754509	KX755181	KX754542	KX754602
<i>Scapanulus oweni</i>	KIZ: 037964	China: Gansu	KX754481	KX754510	KX755182	KX754543	KX754603
<i>Scapanulus oweni</i>	KIZ: 036923	China: Gansu	KX754482	KX754511	KX755183	KX754544	KX754604
<i>Scapanulus oweni</i>	KIZ: 027013	China: Shaanxi	KX754483	KX754512	KX755184	KX754545	KX754605
<i>Parascalops breweri</i>	HT-1	USA: New York	AB076808	KX754496	KX755167	KX754529	KX754589
<i>Scapanus latimanus</i>	BF-1	USA: California	AB076813	KX754497	KX755170	KX754532	KX754592
<i>Scapanus orarius</i>	UWBM:81154	USA: Oregon	KX754472	KX754500	KX755171	KX754533	KX754593
<i>Scapanus orarius</i>	UWBM:81163	USA: California	KX754473	—	KX755172	KX754534	KX754594
<i>Scapanus orarius</i>	CM-3	Canada: British Columbia	AB076816	KX754501	KX755173	—	—
<i>Scapanus townsendii</i>	UWBM:81168	USA: Washington	KX754474	KX754502	KX755174	KX754535	KX754595
<i>Scapanus townsendii</i>	TM-3	USA: Washington	AB076820	KX754503	KX755175	KX754536	KX754596
<i>Scalopus aquaticus</i>	HPW622	USA: Illinois	KX754471	KX754498	KX755169	KX754531	KX754591
<i>Scalopus aquaticus</i>	NMMNH:DJH4822:4348	USA: Iowa	KX754470	KX754497	KX755168	KX754530	KX754590
<i>Euroscaptor longirostris</i>	KIZ: 0905172	China: Sichuan	HG737870	HG737884	KX755148	HG737899	HG737932
<i>Euroscaptor longirostris</i>	KIZ: 0905290	China: Sichuan	HG737871	HG737885	KX755149	HG737900	HG737933
<i>Euroscaptor parvidens</i>	SIK:0859	Viet Nam: Quang Nam	AB823121	AB823156	KX755147	KX754521	HG737931
<i>Scaptorynx fuscicaudus</i>	KIZ: GLGS5532	China: Yunnan	KX754475	KX754504	KX755177	KX754537	KX754597
<i>Scaptorynx fuscicaudus</i>	KIZ: GLGS5974	China: Yunnan	KX754476	KX754505	KX755178	KX754538	KX754598
<i>Parascaptor leucura</i>	KIZ: DY0054	China: Yunnan	HG737877	HG737891	KX755161	HG737905	HG737939
<i>Parascaptor leucura</i>	KIZ: 0412179	China: Yunnan	HG737878	HG737892	KX755162	HG737906	HG737940
<i>Episoriculus caudatus</i>	THUB-S-00005	China: Taiwan	GU981272	GU981030	GU981118	GU981193	GU981451
<i>Uropsilus nivatus</i>	KIZ: 0705081	China: Yunnan	KX754484	KX754513	KX755187	KX754546	KX754606

In addition, we calculated the Kimura-2-parameter (K2P) distances of the complete *CYT B* gene between *Scalopini* sp. and all recognized genera within the *Scalopini* to estimate pairwise genetic distances using MEGA v.7 (Kumar *et al.*, 2016).

RESULTS

MORPHOLOGICAL COMPARISON

The two specimens collected from Mt Namjagbarwa could be assigned to the *Scalopini* based on their dentition. The first upper incisor of these specimens is enlarged, significantly larger than the canine, and is separated from the second upper incisor (Figs 2, 3). This trait is a typical feature of the *Scalopini*, but not in the *Talpini* tribe (Motokawa, 2004). They are further distinguished from all Chinese talpine moles by their relatively longer (more than twice the hind foot length) and hairy tail (Fig. 4).

Using mean greatest length of the skull as an indicator of overall size, *Scalopini* sp. (GLS = 29.08 mm)

is similar to *Scapanulus oweni* (GLS = 29.59 mm), but much smaller than all North American species including *Parascalops breweri* (Bachman, 1842) (GLS = 31.0 mm), *Scapanus latimanus* (GLS = 35.5 mm), *Scapanus orarius* (GLS = 32.1 mm), *Scapanus townsendii* (GLS = 41.2 mm) and *Scalopus aquaticus* (Linnaeus, 1758) (GLS = 32.3 mm) (Table 3; Supporting Information, Table S1). Like in *Scapanulus oweni*, the tail of *Scalopini* sp. (mean, TL = 41 mm) is more than one-quarter the total length (TTL, TL/TTL = 29 %) or twice the hind foot length (TL/HF = 2.27), compared with the relatively shorter tail of *P. breweri*, *Scapanus latimanus*, *Scapanus orarius*, *Scapanus townsendii* and *Scalopus aquaticus*, which is more than one-quarter the total length (Supporting Information, Table S1; Jackson, 1915). The tail of *Scalopini* sp. is moderately covered with pale brown hairs, whereas *Scapanulus oweni* and *P. breweri* tails are densely covered with dark brown hairs. By contrast, in *Scalopus aquaticus* the tail is entirely naked, while that of the *Scapanus* spp. is scantily covered with short silvery hairs. The first toe of the hind foot of *Scalopini* sp. is curved, just like that of *Scapanulus oweni*; this feature clearly distinguishes it from the straight hind foot toes of all the other scalopines. The fore toes and hind toes of most scalopines, including *Scalopini* sp., are not webbed, except for *Scalopus aquaticus*.

The rostrum of *Scalopini* sp. is relatively longer and more slender than in all other species in the *Scalopini*. The zygomatic plates of *Scapanulus* are well developed and obviously larger than those of *Scalopini* sp. and other scalopines. The interorbital region of *Scalopini* sp. is broad and nearly rectangular, similar to that in *Scapanulus oweni*, *P. breweri* and *Scapanus* spp., but relatively broader and more robust. Moreover, the interorbital region of *Scalopus aquaticus* is constricted and narrower than the rostrum. The auditory bullae of *Scalopini* sp., *Scapanulus oweni* and *P. breweri* are incomplete, which is distinct from the complete auditory bullae in *Scalopus aquaticus* and *Scapanus*. The coronoid process of *Scalopini* sp. is reduced and triangular, similar to *Scalopus aquaticus*, but weaker. In *Scapanulus oweni*, *P. breweri* and *Scapanus* spp. the coronoid process is much stronger, with a squared tip.

The dental formula of *Scalopini* sp. (3.1.4.3/3.1.4.3 = 44) is shared with *P. breweri* and *Scapanus* spp. (Fig. 3), but differs from that of the *Scapanulus oweni* (2.1.3.3/2.1.3.3 = 36) and *Scalopus aquaticus* (3.1.3.3/2.0.3.3 = 36). The details of the dental features also differ between *Scalopini* sp. and other species. The unicuspid teeth of *Scalopini* sp. are well spaced, like in most scalopines, with only *Scapanus latimanus* being an exception. In *Scalopini* sp., the I^3 is much smaller than the subequal I^2 , P^1 and P^2 , and the canine is significantly higher than I^2 , P^1 and P^2 . However, the I^2 and I^3 are subequal, and the canines slightly higher in *P. breweri* and *Scapanus* spp. In *Scapanulus oweni*,

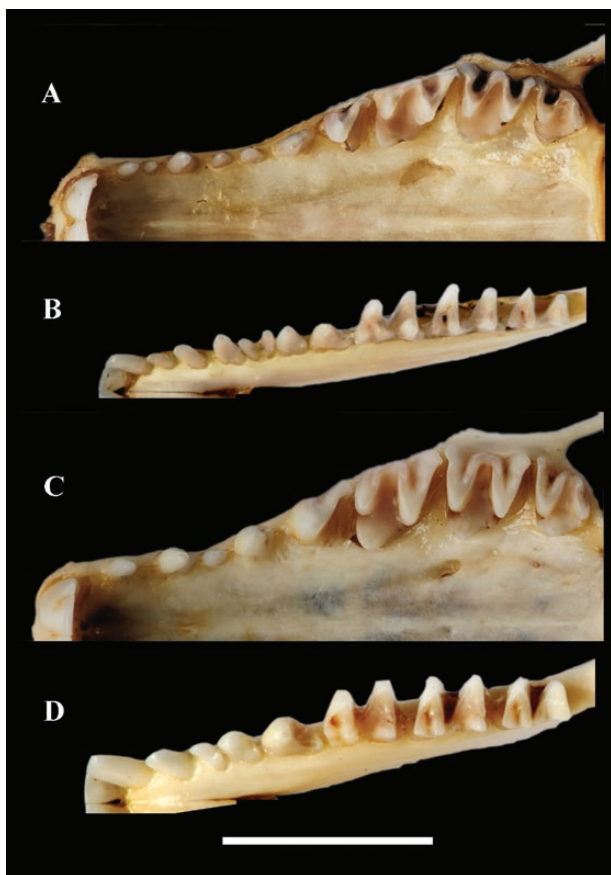


Figure 3. Occlusal views of the right lower teeth and upper teeth of *Alpiscaptulus medogensis* (KIZ: 037966; A, B) and *Scapanulus oweni* (KIZ: 033872; C, D). Scale bar = 5 mm.



Figure 4. Skins (dorsal view, left and ventral views, right) of the holotype of *Alpiscaptulus medogensis* (KIZ: 037966).

only two incisors are present, and P^1 is much smaller than I^2 and P^2 (Fig. 3), while in *S. aquaticus*, I^2 and I^3 are tiny and P^1 is smaller than P^2 . The M^1 and M^2 of Scalopini sp., *Scapanulus oweni* and *P. breweri* are broad with three lobes, instead of narrow, simple upper molars with only a V-shaped cusp-like shelf (not lobed) like in *Scapanus* and *Scalopus*.

PHYLOGENY AND MOLECULAR DATING

We obtained 4043 bp sequences for each specimen, including 1961 bp mitochondrial [*CYT B* (1140 bp) and 12S *rRNA* (821 bp)] and 2082 bp nuclear [*APOB*

(579 bp), *BRCA1* (810 bp) and *RAG2* (693 bp)] sequences. All new sequences have been deposited in GenBank under Accession Nos MT349878–879 and MT349295–302 (Table 2). The K2P distances of *CYT B* showed large genetic distances between Scalopini sp. and all recognized Scalopini genera, ranging from 14.5% to 18.9% (i.e., 14.5%, 16.1%, 17.4% and 18.9% against *Parascalops*, *Scapanulus*, *Scapanus* and *Scalopus*, respectively, Table 4).

The maximum likelihood (ML) and Bayesian phylogenetic reconstructions recovered identical topologies, and thus only the ML trees are shown (Fig. 5). The two new specimens were unambiguously embedded

Table 3. Weight (in grams) and external and craniomandibular measurements (in mm) of two specimens of *Alpiscaptulus medogensis* and *Scapanulus oweni* (means $\pm$ standard deviation, observed ranges, and number of specimens)

	<i>Alpiscaptulus medogensis</i>		<i>Scapanulus oweni</i>
	KIZ: 037966	KIZ: 037965	N = 6
	♀	♂	
W	25.9	34.8	32.34 $\pm$ 7.63 20.50–41.80; 5
HB	89	111	100.83 $\pm$ 13.17 80.00–116.00; 6
TL	42	40	40.00 $\pm$ 6.51 33.00–49.00; 6
HF	18	18	16.17 $\pm$ 1.83 14.00–19.00; 6
GLS	29.08		29.59 $\pm$ 0.96 28.45–30.73; 4
BL	24.67		25.23 $\pm$ 0.65 24.46–26.03; 4
PL	12.83	12.71	13.40 $\pm$ 0.43 12.90–13.97; 5
CH	8.33		8.56 $\pm$ 0.04 8.52–8.60; 3
CB	13.41		13.77 $\pm$ 0.40 13.34–14.30; 4
IOB	6.67	6.97	6.53 $\pm$ 0.19 6.36–6.80; 4
ZB	9.85	9.5	10.77 $\pm$ 0.20 10.53–11.03; 4
UTL	12.43	11.88	12.86 $\pm$ 0.34 12.42–13.25; 5
UML	5.05	4.85	5.48 $\pm$ 0.10 5.41–5.62; 4
M ² -M ²	7.5	7.68	8.39 $\pm$ 0.22 8.12–8.73; 5
P ⁴ -M ³	6.18	6.39	7.04 $\pm$ 0.16 6.95–7.28; 4
LTR	11.17	10.9	11.47 $\pm$ 0.30 11.10–11.82; 4
HCP	6.36	7.24	7.27 $\pm$ 0.33 6.88–7.67; 4
HCV	4.57	4.25	4.22 $\pm$ 0.20 3.93–4.41; 4

within the clade Scalopini, and a sister relationship between Scalopini sp. and *Scapanulus oweni* was strongly supported in all analyses (UFboot > 97, PP = 1.00). Two major clades in the Scalopini (i.e., Scalopini sp.+*Scapanulus*+*Parascalops* and *Scapanus*+*Scalopus*) are strongly supported in the nuclear and mitochondrial-nuclear combined gene trees (UFboot = 100, PP = 1.00);

Table 4. The K2P distance of the *CYT B* gene in the genera of the Scalopini

	<i>Alpiscaptulus</i>	<i>Parascalops</i>	<i>Scapanulus</i>	<i>Scapanus</i>
<i>Alpiscaptulus</i>				
<i>Parascalops</i>	0.145			
<i>Scapanulus</i>	0.161	0.171		
<i>Scapanus</i>	0.174	0.156	0.189	
<i>Scalopus</i>	0.189	0.186	0.217	0.172

however, the support value was low in the mtDNA tree (UFboot = 41, PP = 0.78). All analyses inferred a sister relationship between *Parascalops breweri* and the Chinese lineage (i.e., Scalopini sp.+*Scapanulus oweni*); however, this was supported only in the Bayesian analysis of mitochondrial-nuclear combined gene trees (PP = 0.97). The BEAST divergence analyses indicated that the two major clades of the Scalopini diverged from each other at about 21.37 Myr (95% CI = 17.26–25.65 Mya), while Chinese and North American scalopine moles diverged at about 17.64 Myr (95% CI = 14.93–20.27 Mya). Scalopini sp. and *Scapanulus oweni* diverged in the mid-Miocene at about 11.56 Myr (95% CI = 8.28–14.79 Mya) (Fig. 6).

Based on the distinct morphological characters and deep genetic divergence described in the preceding sections, we recognize the two moles from Mt Namjagbarwa as representing a distinct, undescribed genus and species within the Scalopini, and formally describe it as follows.

FAMILY TALPIDAE G. FISCHER, 1814

SUBFAMILY TALPINAE G. FISCHER, 1814

TRIBE SCALOPINI T.P. GILL, 1875

ALPISCAPTULUS CHEN & JIANG, GEN. NOV.

Zoobank registration: urn:lsid:zoobank.org:act:475D09CD-EB46-4522-ACE6-733C983CAA86

Type species: *Alpiscaptulus medogensis* Jiang & Chen, sp. nov.

Diagnosis: The following combination of characters defines the new genus. Medium in size (HB = 100 mm; GLS = 29.08 mm). The rostrum is long and slender. The tail is more than one-quarter the total length or twice the length of the hind foot and is moderately covered with pale brown to white hairs (Fig. 4). The chin is covered with white hairs. Fore toes and hind toes are not webbed, and the first toe of hind foot is curved. The dental formula is 3.1.4.3/3.1.4.3 = 44. The zygomatic plates are less developed. The interorbital region is broad, without constriction and appears rectangular. The pterygoid region and the coronoid valley are

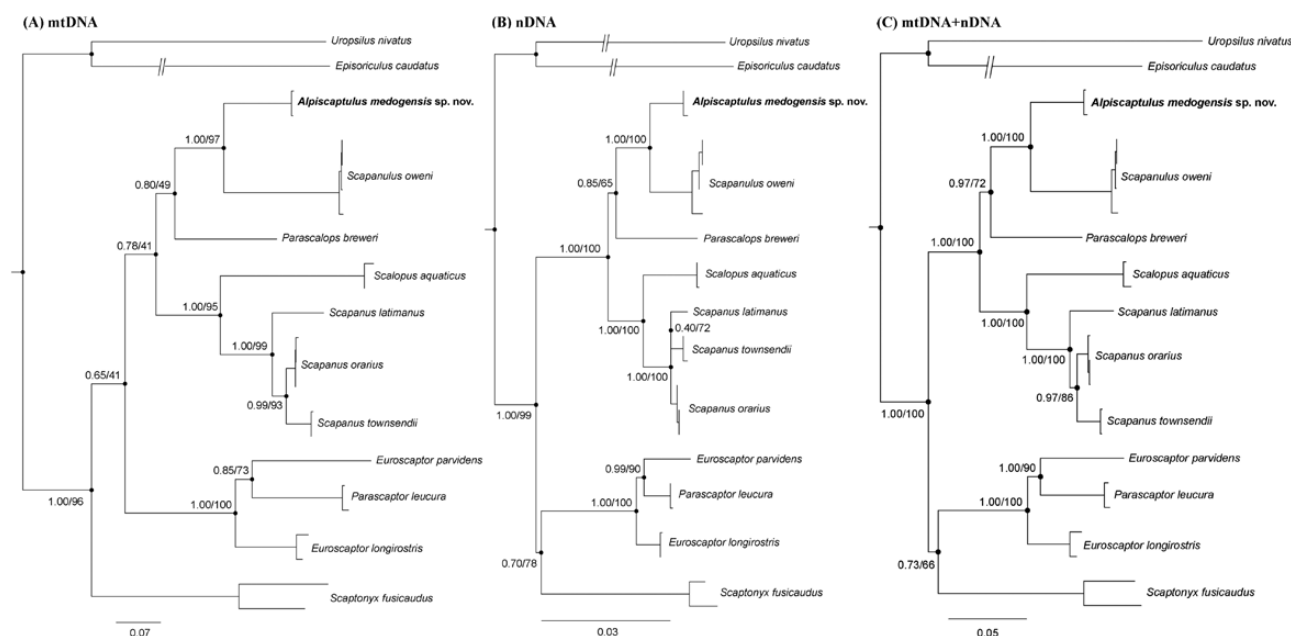


Figure 5. Results of maximum likelihood phylogenetic analyses of concatenated (A) mitochondrial genes, (B) nuclear genes and (C) mitochondrial-nuclear trees. Node numbers indicate Bayesian posterior probabilities (PP) and ultrafast bootstrap supports (UFBoot). Branch lengths represent substitutions per site.

shallow, with a weak and triangular coronoid process. The auditory bullae are incomplete.

Description: Because the new genus is monotypic, the description is the same as in the species description, presented below.

Etymology: The genus name *Alpiscaptulus* is derived from the Greek $\alpha\lambda\pi$, alp or high mountain, and $\sigma\kappa\alpha\pi\tau\acute{o}\lambda o\varsigma$, a small digger. The genus name refers to a small digging animal living in the mountains.

Comparisons: Among the Scalopini tribe, *Alpiscaptulus* is most similar to the monotypic genus *Scapanulus*, and they are also the only two genera distributed in Asia. The two species have similar body sizes and share some unique characters in contrast with other species, such as curved thumbs of hind feet and relatively long tail, which is more than one-quarter the total length. However, they are strikingly dissimilar in many other external characters. The tail of *Alpiscaptulus* is moderately covered with pale brown to white hairs, unlike the dense dark brown hairs in *Scapanulus*. Thus, the tail of *Alpiscaptulus* is more slender than *Scapanulus* in appearance. The rostrum of *Alpiscaptulus* is much longer and more slender than that of *Scapanulus*. The most distinctive cranial difference between *Alpiscaptulus* and *Scapanulus* is the dentition. The possession of 44 teeth in *Alpiscaptulus* is distinguishable from *Scapanulus* (36 teeth), and the teeth of *Alpiscaptulus* are weaker and smaller than in *Scapanulus*. It can further be

distinguished from *Scapanulus* by the less developed zygomatic plates, broader interorbital region, much shallower pterygoid region and coronoid valley, the reduced, triangular coronoid process and the relatively smaller measurements of UTL, M^2 - M^2 , UML, P^4M^3 , ZB and CH (Table 3).

Alpiscaptulus can be distinguished from the North American *Scalopus*, *Scapanus* and *Parascalops* by its smaller size, more slender rostrum, curved thumbs of hind feet and by the relatively longer (more than one-quarter the total length), hairy and pale brown tail. Furthermore, the reduced and triangular coronoid process is much weaker than that of *Parascalops* and *Scapanus*, while the incomplete auditory bullae of *Alpiscaptulus* is noticeably different from *Scalopus* and *Scapanus*. The 44 teeth of *Alpiscaptulus* are distinct from *Scalopus* (36 teeth) and the webbed fore toes and hind toes of *Scalopus* are unique among scalopines.

ALPISCAPTULUS MEDOGENSIS JIANG & CHEN, SP. NOV.

Suggested common name: Medog mole, 墨脱鼹 (motuo yan).

Zoobank registration: urn:lsid:zoobank.org:act:292B5D9C-4F54-4C58-B0F8-2E7BF7DE0D1D

Holotype: KIZ: 037966, an adult female collected by Kang Luo at on 2 June 2019. Dried skin, cleaned skull and alcohol-preserved carcass.

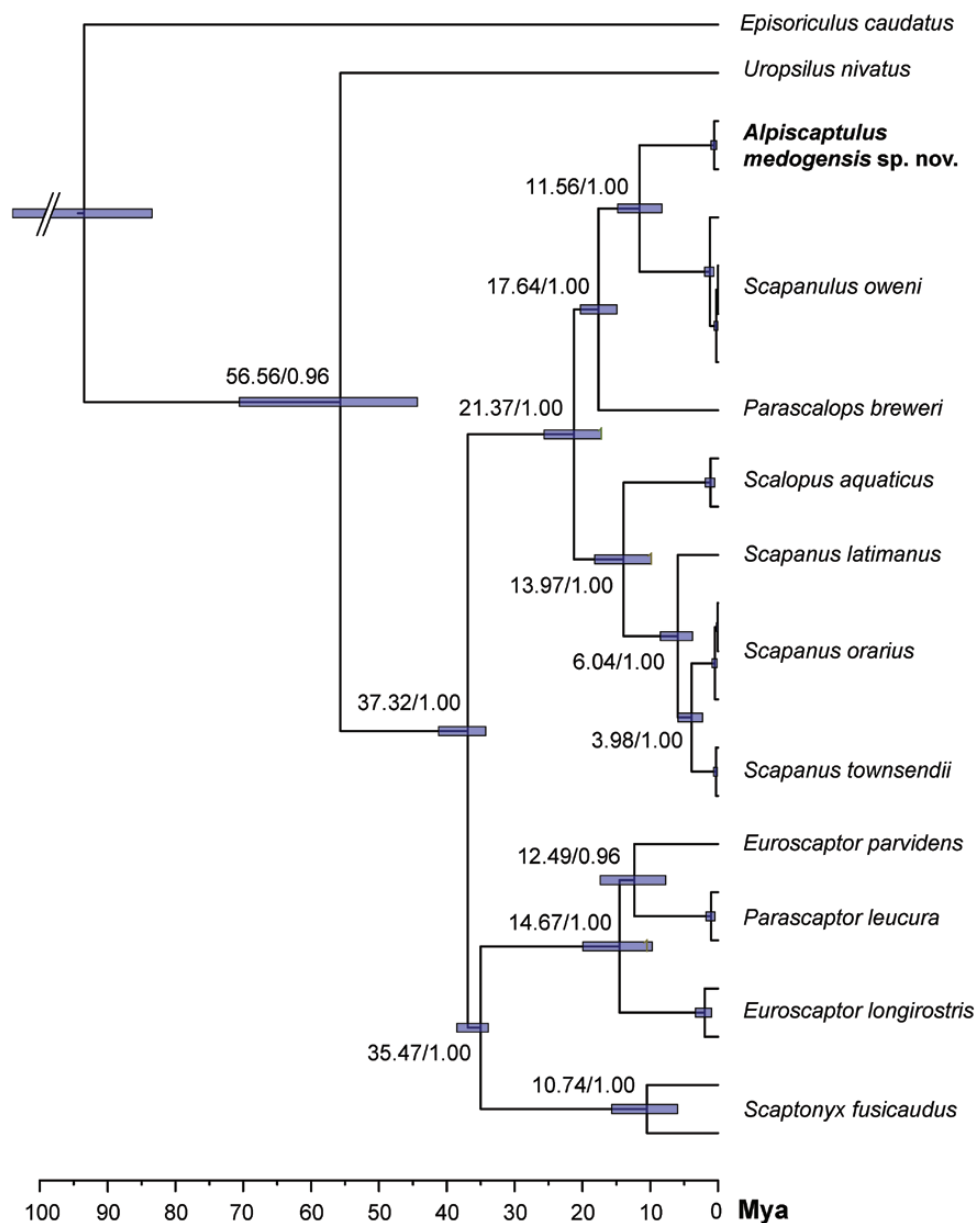


Figure 6. Divergence times estimated using BEAST based on mitochondrial-nuclear concatenated data. Node numbers refer to divergence time in million years (Mya) and Bayesian posterior probabilities (PP).

Type locality: Mt Namjagbarwa, Damu Town, Medog County, south-east Tibet, China (29°44'32" N, 95°40'59" E), elevation approximately 3650 m a.s.l.

Paratype: KIZ: 037965, an adult male collected by Changzhe Pu on 24 May 2019 at an elevation of 2400 m a.s.l. from Gedang Town, Medog (29°29'31" N, 95°45'11" E). Dried skin, cleaned skull (braincase damaged) and alcohol-preserved carcass.

Measurements: See Table 3.

Diagnosis: As for genus (see above).

Description: A medium-sized mole (HB = 100 mm, TL = 41 mm, HF = 18 mm). Dorsal pelage dark grey, ventral pelage imperceptibly lighter. Eyes minute, concealed in fur; external ears absent. The nose is long and develops into a conical snout; the nose and chin are covered in white hairs (Fig. 4). The tail is about twice the length of the hind foot and is moderately covered in pale brown to white hairs, with a tuft of longer hairs located on the tip. The dorsal surfaces of hands and feet are light brown, white at the margin. Fore toes and hind toes not webbed, and the palms are nearly equal in length and width, with the flattened

and slender claws. The hind foot is long and narrow, and the first toe is curved like that of *Scapanulus oweni*, which is set outward at a slight angle against remaining toes. We observed the curved second toe of the right hind foot in specimen KIZ: 037965, but not in specimen KIZ: 037966.

The skull is triangular and flat (Fig. 2). The rostrum is long and slender; the braincase is broad with well-developed mastoid processes. The zygomatic arch is moderately heavy, and slightly convergent anteriorly; zygomatic plates are small. The orbits are narrow and situated in the middle of the skull. The interorbital region is broad, strong, and nearly rectangular. The infraorbital foramen is small. The pterygoid region is shallow, and auditory bullae are incomplete. The coronoid process is weak and triangular; the condyloid process is short and angled upward at roughly 45°, so the coronoid valley appears shallow. The angular process is well developed, curved upward at the posterior.

The dental formula is I 3/3, C 1/1, P 4/4, M 3/3, totalling 44, and the incisors, canines and premolars are all well-spaced. The I¹ is enlarged and flat and twice the height of I², while the I³ is the smallest of all. The canine is larger than I² and I³, P¹ and P² are subequal and smaller than the canine, P³ is almost as high as canine but broader, and P⁴ is much larger than P³. The transverse section of the upper molar is W-shaped, M¹ > M² > M³; M¹ and M² broad with a trilobed basal shelf, while M³ is much smaller and bilobed. I₁ and I₂ are enlarged and flattened, and I₃ is much smaller. The canine is slightly larger than I₃. All three incisors and the canine lean forward. All premolars are subequal; the P₂ of the right side of the type specimen has two equal tooth tips (Figs 2, 3), but only one tip on the other specimen. M₁ and M₂ are almost equal, M₃ is slightly smaller.

Etymology: The species name *medogensis* derives from Medog, the type locality of the new species, and the Latin adjectival suffix *-ensis*, meaning “belonging to”.

Comparisons: Because all genera of the Scalopini except *Scapanus* are monotypic, the comparisons are the same as the genus comparisons presented above.

Distribution: *Alpiscaptulus medogensis* is only known from the type locality at Mt Namjagbarwa in Medog County, south-east Tibet, China. The known elevational range is 2400 m a.s.l. to 3700 m a.s.l..

Ecology and habitat: From 20 April to 3 June 2019, we sampled small mammals along the elevational gradients from 650 to 3800 m a.s.l. at Mt Namjagbarwa. The two specimens of *A. medogensis* were captured at relatively high elevations, 2400 m a.s.l. and 3700 m

a.s.l., respectively. During the survey, we did not see mole runways and mounds of excavated earth in this area. At 2400 m a.s.l., the mole was captured in a small hole using a mole trap. The habitat at this area was brushy, dominated by oak forest, with abundant grass cover on the ground. The mole from 3700 m a.s.l. was captured by a plastic bucket pitfall (15 cm in diameter and 20 cm in height). The vegetation of this area was dominated by heath forest, with mixed bamboo forests.

DISCUSSION

Both the morphological and molecular analyses strongly suggest that the two specimens collected from Mt Namjagbarwa represent a new taxon in the Scalopini. It is diagnosably distinct from all other genera of the Scalopini by the combination of the pale brown tail, curved thumb and possession of 44 teeth. Additionally, the ancient divergence (c. 11.59 Mya) and large genetic distances from other genera (14.5–18.9% by the *CYT B* gene) recognize it as a distinct genus. The discovery of a new scalopine genus in the Himalayas is unexpected and of great interest because the distribution of most species in this group is limited to North America (Hutterer, 2005). Thus far, only two species (i.e., *Scapanulus oweni* and the above-described *A. medogensis*) are found in China, and the new genus occurs even farther west, in the eastern Himalayas. It is noted that several characters in *A. medogensis*, such as the possession of 44 teeth, the less developed zygomatic plates and the shallower pterygoid region, are more similar to the North American *P. breweri* than Chinese *Scapanulus oweni*. Also, although only two specimens were collected, some unstable traits were observed (i.e., the curved second toe of the right hind foot of the *Paratype*; two equal tooth tips on the P₂ of the type specimen, Figs 2, 3). This might be due to the relatively stable habitats in the eastern Himalayas, which allowed *A. medogensis* to retain more ancestral traits than *Scapanulus oweni*.

Most of the clades within the Scalopini were supported as monophyletic in our phylogenetic trees (UFboot ≥ 95, PP ≥ 0.95; Fig. 5). The new species *A. medogensis* was unambiguously embedded within the Scalopini, and the sister relationship with *Scapanulus oweni* was strongly supported (UFboot > 97, PP = 1.00). Hutchison (1968) divided the Scalopini into two subtribes (i.e., Scalopina and Parascalopina) based on morphological characters. The Scalopina includes the extant *Scalopus* and *Scapanus*, and the fossil taxon *Scapanoscapter* Hutchison, 1968; while the Parascalopina includes the extant *Parascalops* and *Scapanulus*, and two fossil taxa *Domninoidea* Green, 1956 and *Mioscalops* Ostrander *et al.*, 1960. Our research does not include fossil

species; however, two major clades in the Scalopini had significant support in the nuclear and mitochondrial-nuclear combined gene trees (UFboot = 100, PP = 1.00; Fig. 5), one consisting of *Alpiscaptulus*, *Scapanulus* and *Parascalops*, and the other consisting of *Scalopus* and *Scapanus*. The two subtribes had a deep genetic divergence from each other at about 21.37 Myr (95% CI = 17.26–25.65 Mya) (Fig. 6). Such a result is consistent with the categories of Hutchison (1968) of the extant species, with *Alpiscaptulus* belonging to the subtribe of *Parascalopina*.

As the second relict species and the most south-westerly distributed species of the Chinese scalopines, the discovery of *A. medogensis* is important in understanding the biogeographical history of the Scalopini. The divergence time of *A. medogensis* and *Scapanulus oweni* is about 11.56 Myr (95% CI = 8.28–14.79 Mya), which corresponds to the rapid diversification of Asian mole genera (He et al., 2014). During this period, the Himalayan-Tibetan Plateau experienced a rapid uplift, which enhanced global aridity and marked cooling (Coleman & Hodges, 1995; Zachos et al., 2001; Sun et al., 2005). Thus, we suggest that the rapid uplift of the Himalayan-Tibetan Plateau and ensuing climate change led to the isolation and divergence of *A. medogensis* and *Scapanulus oweni*. Schwermann et al. (2019) hypothesized that scalopine moles migrated from North America into Asia at least twice via the Bering land bridge, and the earlier time (c. 15–16 Mya) represents the origin of *Scapanulus oweni*. According to this scenario, the ancestor of *A. medogensis* and *Scapanulus oweni* may have migrated via central China and colonized the Himalayas in the mid-Miocene and then became isolated and diverged owing to the rapid uplift of the Himalayan-Tibetan Plateau over a relatively short period of time. Most scalopine species in China are now extinct; however, *A. medogensis* may have persisted because the region where it currently occurs potentially acted as a refugium during glacial and interglacial periods (Lei et al., 2014). There is now a major disjunction between the distributions of *A. medogensis* and *Scapanulus oweni*, largely owing to the Salween, Mekong and Yangtze rivers. Considering that such rivers frequently play an important role in speciation of small mammals (He et al., 2014; Chen et al., 2017), other scalopine moles are likely to be found in the Himalayas and Hengduan Mountains.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Table S1. Means and observed ranges of external and selected cranial measurements of American-distributed Scalopini from previous studies.